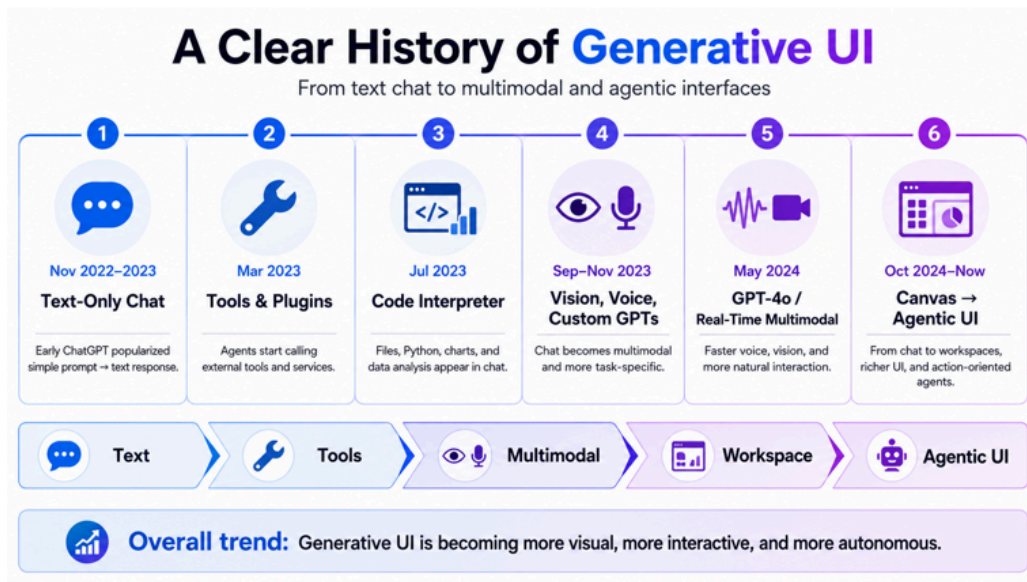
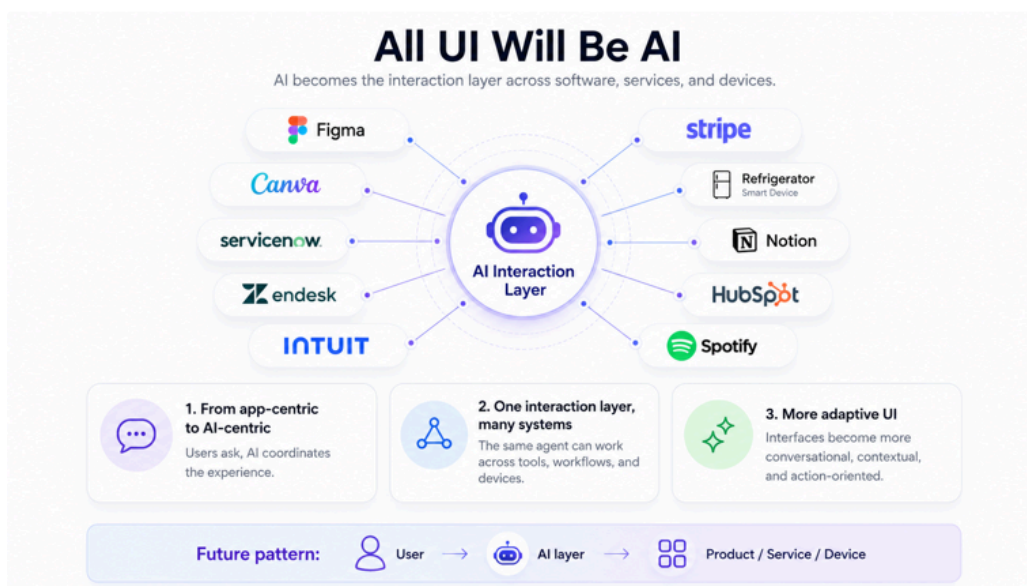


🧩 The future of UI: A2UI

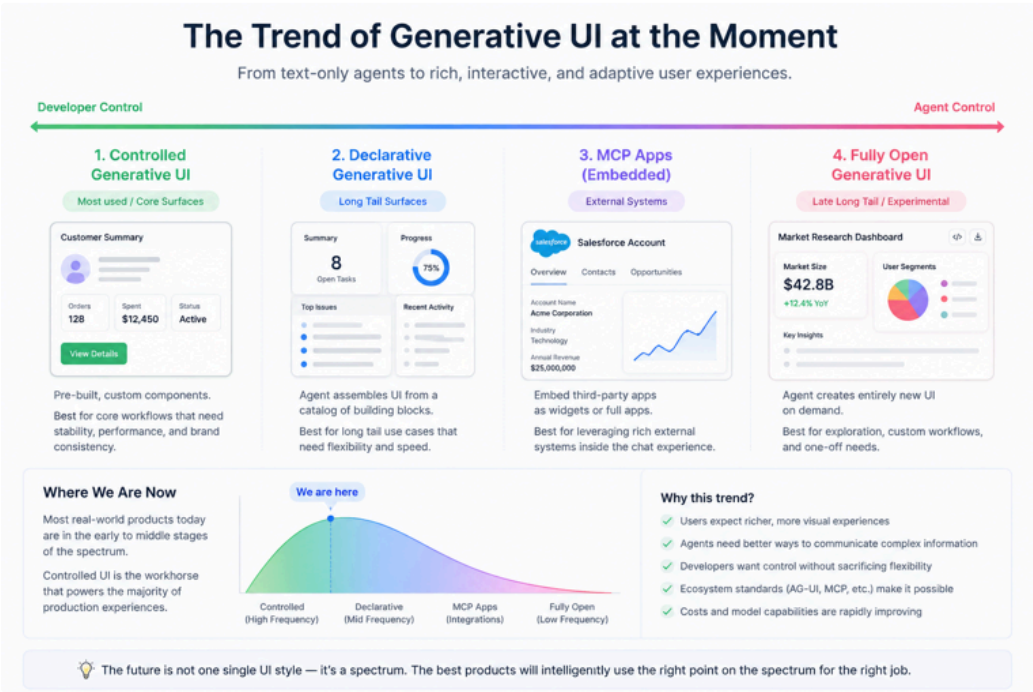
Background



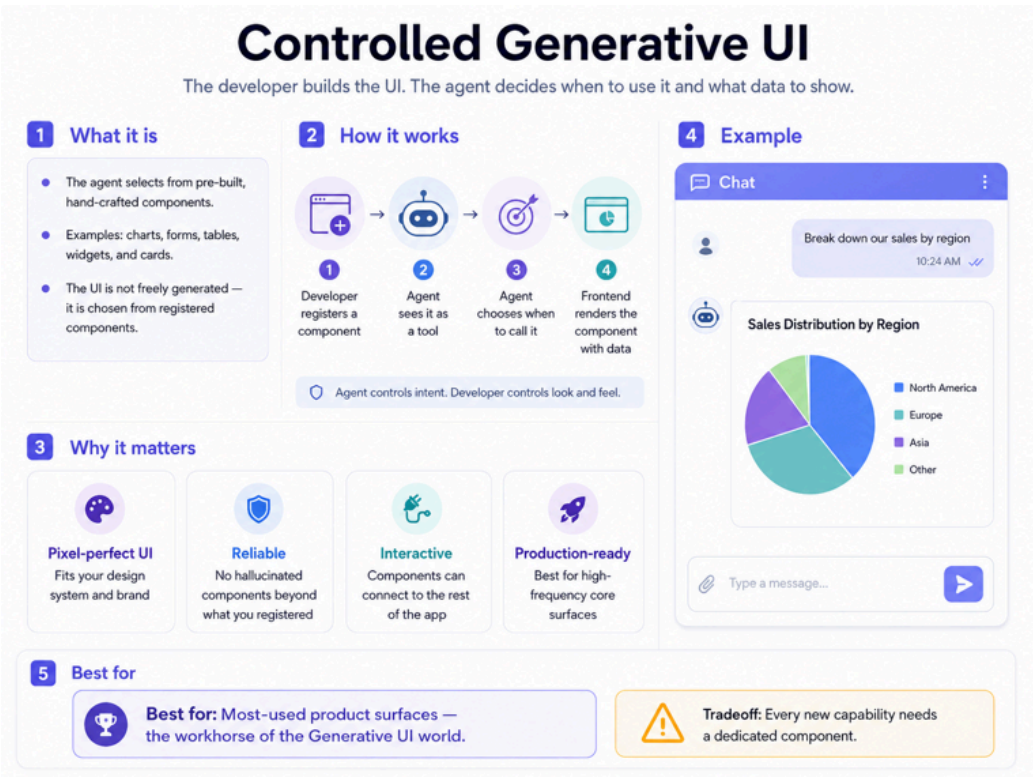
Where we are heading to



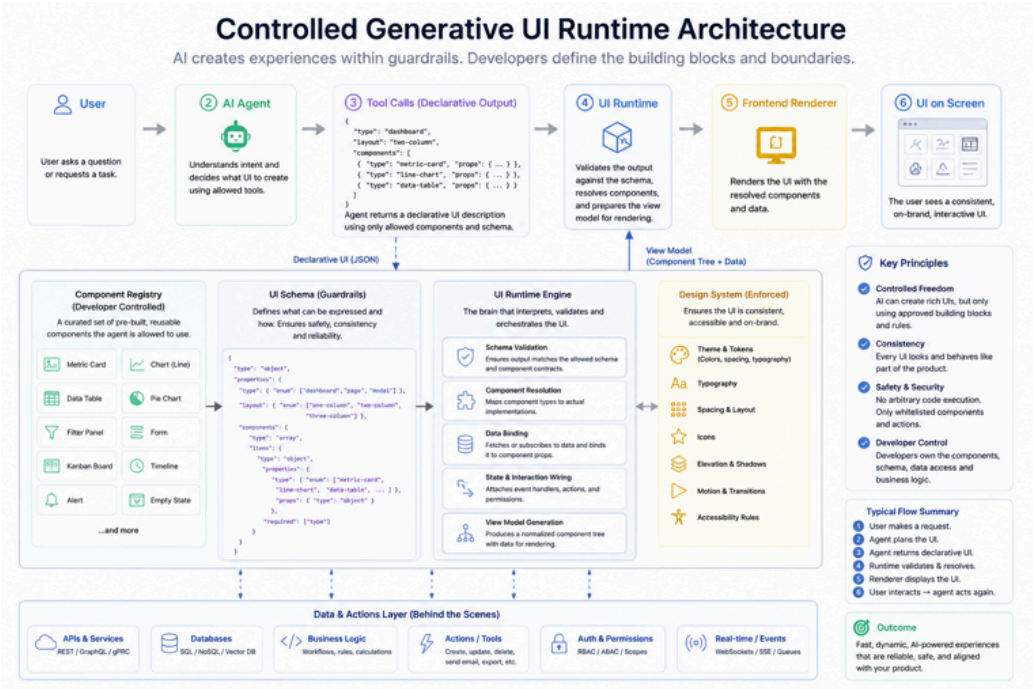
The trend of Generative AI UI



The Controlled Generative UI



The UI Runtime Architecture



From Controlled Generative UI to Declarative UI

Controlled Generative UI

Pre-Built Custom Components



Core Idea

Developers define a set of custom UI components the agent is allowed to use — like giving the agent a toolbox. The agent selects from those components instead of freely generating any interface.



Attributes

- ✓ Most familiar pattern
- ⚡ Easiest to build
- 🛡️ Highest developer control
- ⚙️ Requires upfront component work



Freedom of Expression

1

Controlled UI
Predefined components

2

Declarative UI
Structured schema

3

Open-Ended UI
Free-form generation



What can be represented? From fixed UI to fully flexible UI.



Who Has Control?



Agent

~70-80%



Programmer

In controlled UI, the developer constrains the interface while the agent decides when to use the available components.



Takeaway

Controlled Generative UI offers the most control and reliability. It is ideal for core product surfaces, but developers must define the component library upfront.

The interesting part:

Frontend Tools Explained

Register frontend functions (components) the agent can call — just like any backend tool.

What is a Frontend Tool?

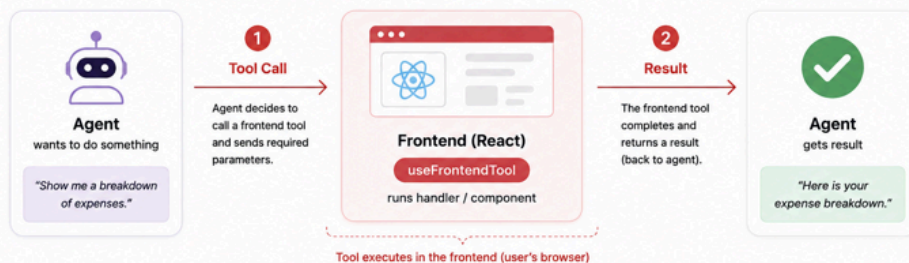


A frontend tool is a UI component or function (e.g., a React component) registered with the system. The agent can call it like a tool. The tool runs in the user's browser (frontend) and can return a result back to the agent.

Examples

- Show a pie chart
- Display a flight card
- Open a modal to get user input
- Render a table, form, or dashboard

Frontend Tool Call Flow



Key Takeaway



Agent decides to call a tool



Frontend tool runs in the browser



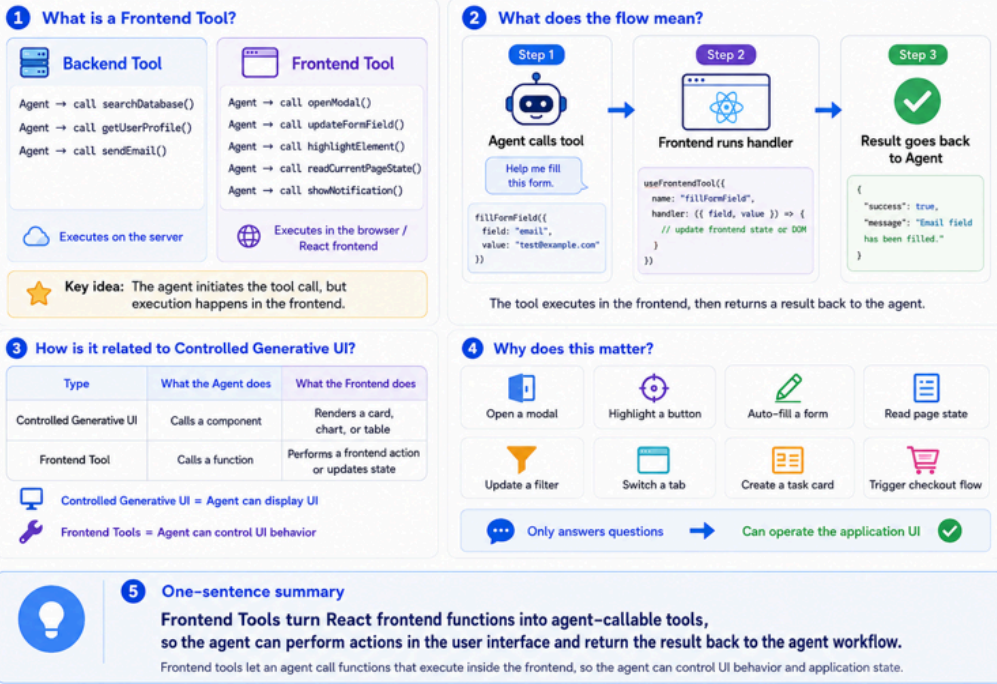
Tool returns result (data or action outcome)



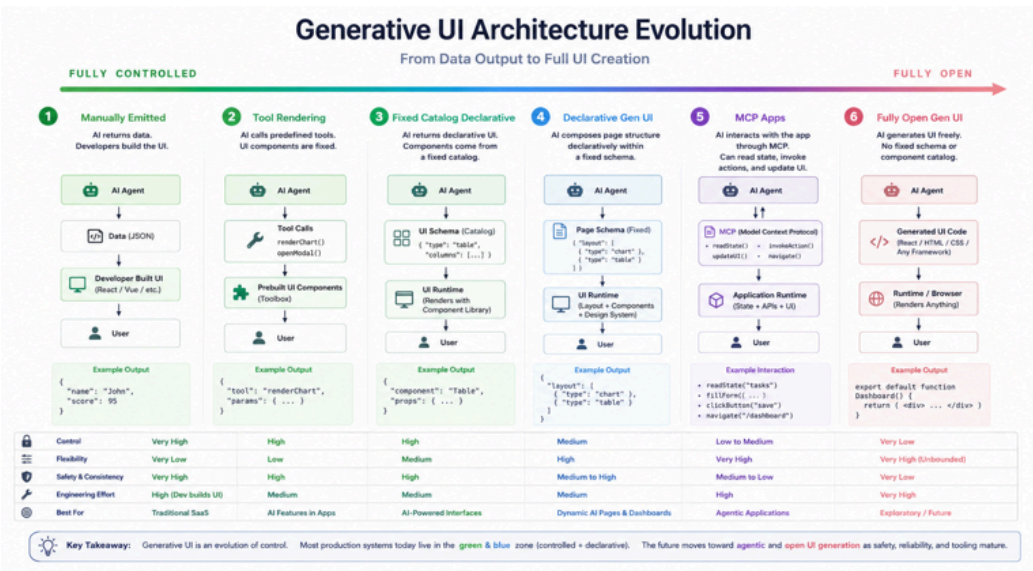
Agent receives the result and continues

Frontend Tools

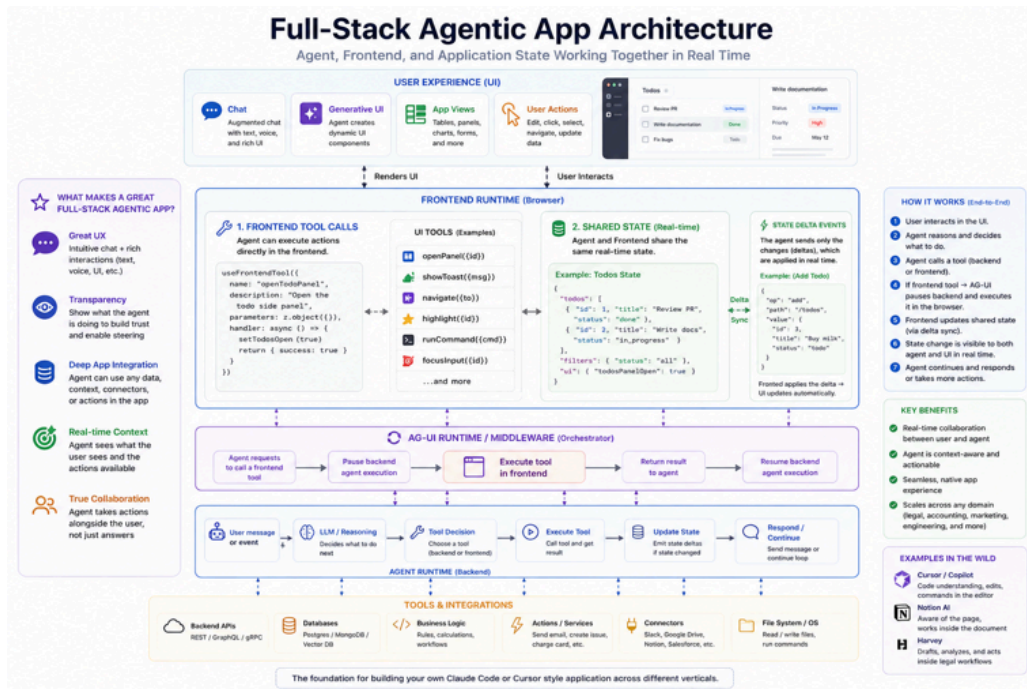
Register frontend functions that an agent can call — so the agent can act inside the UI, not just talk about it.



The Generative UI Architecture Evolution



The full stack Agentic APP Architecture



My Takeaway:

- **Chat** is the entry point.
- **UI** is the workspace.
- **Tools** are the hands.
- **State** is the memory.
- **Human** approval is the safety layer.

Reference

1. [Create | A2UI Composer](#)
2. [A2UI](#) (Google)
3. [CopilotKit | Bring Users and AI Agents together inside real apps](#)
4. [AI SDK](#)

